

Mackenzie Ward

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Incoming Natural Sciences (Physical) student, Pembroke College, University of Cambridge

Profile

Mathematically driven incoming Natural Sciences student with interests in quantum mechanics, computational physics, quantum computing and materials science. Experience in independent academic work, programming, teaching and collaborative technical projects, with particular interest in applying mathematical and computational methods to physical problems.

Education

Pembroke College, University of Cambridge — Natural Sciences		<i>Oct 2026 – Present</i>
Incoming undergraduate in Physical Natural Sciences		
Sir Henry Floyd Grammar School — A-levels	A* A* A* A	<i>Sep 2024 – Aug 2026</i>
Physics, Chemistry, Mathematics, Further Mathematics		
Extended Project Qualification	A*	<i>Sep 2024 – Aug 2025</i>
<i>The NISQ Era and Beyond: Will Quantum Computing Have a Net Positive Effect on the Future?</i>		
Sir Henry Floyd Grammar School — GCSEs/Level 2 qualifications	9999988877	<i>Sep 2019 – Aug 2024</i>
Including 9s in Mathematics and Further Mathematics		

Academic Enrichment

Quantum Club — University of Oxford		<i>Sep 2025 – Present</i>
Weekly quantum-physics tutorials; independent reading of Lvovsky's <i>Quantum Physics: An Introduction Based on Photons</i> ; monthly assignments		
COMPOS Year 12/13 — University of Oxford		<i>Sep 2024 – May 2026</i>
Regular mathematics and physics assignments and weekly tutorial sessions		
UNIQ Engineering — University of Oxford		<i>Jul 2025 – Nov 2025</i>
Selected for the summer school residential; attended engineering lectures/workshops and completed work on differential equations		
Oxford Physics Summer School — University of Oxford		<i>Jul 2025 – Aug 2025</i>
Quantum-physics lectures and poster project, <i>Simulating Spin-Field Interactions in Simple Molecules for Quantum Circuits</i> , completed with support from a PhD candidate		
Theory of Knowledge Enrichment		<i>Sep 2024 – Jun 2025</i>
Studied epistemology; wrote a paper on differences in mathematical understanding despite shared learning environments, published in the school journal		

Technical Projects & Skills

- **Quantum computing:** Python/Qiskit simulations of Bell states, quantum teleportation and GHZ-state non-locality.
- **Programming:** Python, Qiskit, Jupyter; JavaScript, HTML/CSS, Scratch, Roblox Lua and Unity.
- **Numerical methods:** Implemented Euler, midpoint/RK2 and RK4 methods in Python for differential equations.
- **Projects:** Poker AI, programming/electronics projects, custom mechanical keyboard and Raspberry Pi portable-computing setup.
- **Core skills:** Mathematical reasoning, problem solving, technical communication, teaching, teamwork and time management.

Work Experience

Code Sensei — Code Ninjas Aylesbury	<i>Jul 2024 – Present</i>
Teaching 7–15 year-olds through 1–2 hour coding sessions covering Scratch, JavaScript, HTML/CSS, Micro:Bit, Roblox Lua and Unity. Supported in camps, delivering one-to-one taster sessions and helped train new staff.	
Kitchen Porter — The Rising Sun & Pad Thai Cuisine	<i>Sep 2023 – Mar 2025</i>
Worked reliably in a fast-paced kitchen, maintaining hygiene, processing dishes efficiently and supporting food preparation and the wider team during busy services.	
Volunteer — Oxfam, Thame	<i>Jan 2023 – Feb 2025</i>
Operated the till, supported customers, restocked and priced items, and assessed donated goods for quality and suitability.	

Awards & Activities

- **The Mr Lewin Mathematical Critical Thinking Award (2026):** Inaugural school recipient awarded for mathematical accuracy and problem-solving.
- **British Physics Olympiad (BPhO) Round 1 Gold Award (2025); BPhO SPC Bronze Award (2025); BPhO Round 1 Bronze II Award (2024).**
- **UK Chemistry Olympiad Round 1 Gold Award (2025, 2026)**
- **Q-CTRL Quantum Professional Course — Feb 2025.**
- **Duke of Edinburgh Gold Award — in progress;** Silver and Bronze Awards completed.